

Neural Networks and Fuzzy Systems

Introduction
CSE 0613 4221

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Introduction

- ◆ Human and Computer:
 - **Human beings are more intelligent than computers.** Why is that said?
 - Calculating the sum of a few hundred eight and nine digit numbers
 - Does that make the computer more intelligent than us?
 - Consider: **crosswords**
 - Some of us are excellent
 - difficult for a computer
 - Computers are very bad at performing simple **visual tasks**.

Introduction

- ◆ Human and Computer:
 - After 30 years of high-quality research, are **not** sufficient to allow them to make the claim that they have computer systems that are artificially intelligent in any general sense that we would recognize.
 - The **aim** of artificial intelligence could be summed up as trying to make computers behave as they do in the **movies**.

Introduction

- ◆ Human and Computer:
 - Why then can't computers do the things that we do?
 - **The nature of their design**
 - **Brain:** slow, highly parallel; **Computer:** high-speed, serial machine
 - Counting is an essentially serial activity(Computer winner)
 - **vision or speech recognition (Brain Winner)**
- ◆ [Note: Computers are designed to carry out one instruction after another, extremely rapidly, whereas our **brains work with many more slower units**. Whereas a computer can typically carry out **a few million operations** every second, the units in the brain respond about **ten times per second**.]

Introduction

- ◆ For Computer:
 - **Represent Complex information**
 - **Design Learning Algorithm**
 - **Find the technique of parallel processing**

The Structure Of The Brain

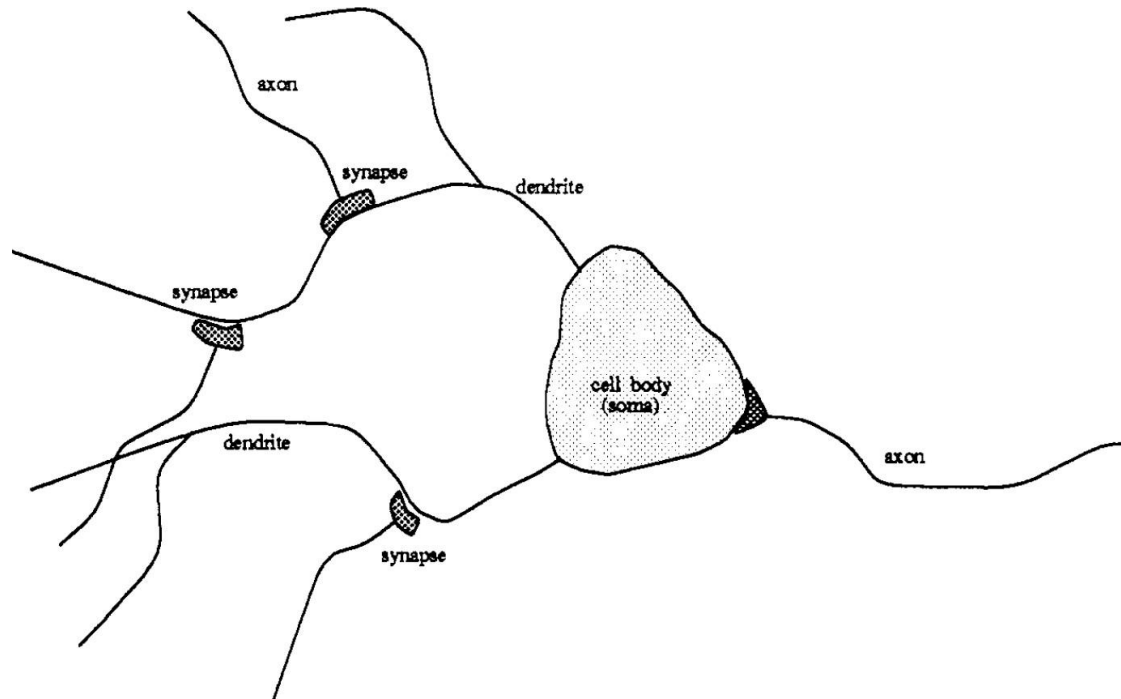


Figure 1.1 The basic features of a biological neuron.

The human brain is very complex and not fully understood, but we know it has about **10 billion neurons**, each linked to around **10,000 others**. Neurons are tiny processing units that receive signals through **dendrites**. If enough input is received, the neuron **fires** a signal. There are two types: some connect nearby neurons, and others link different brain or body parts.

The Structure Of The Brain

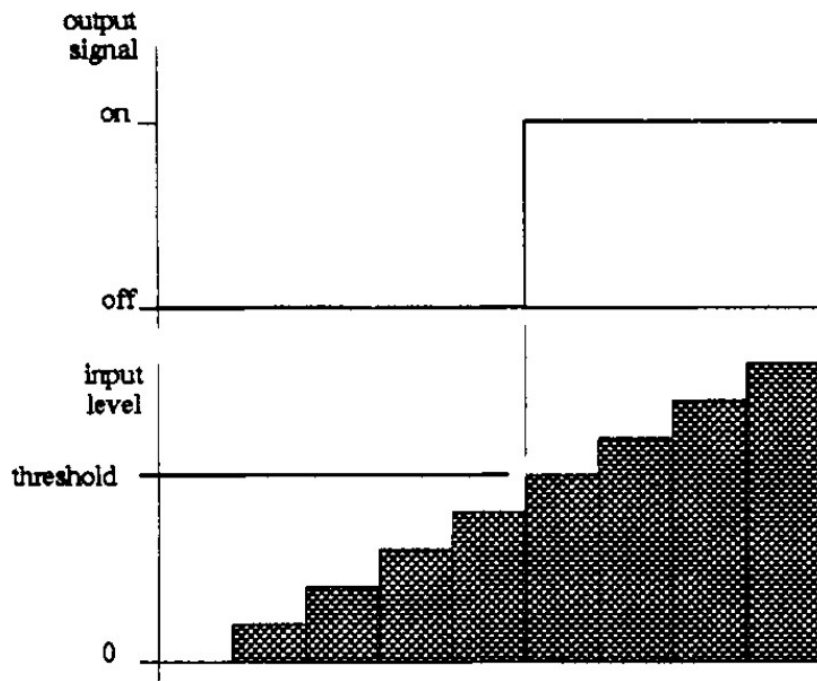


Fig: The input to the neuron body must exceed a certain threshold before the cell will fire.

The image shows how a neuron works using a threshold.

As the input increases, the neuron stays "off" until the input above a certain level goes threshold).

Once the input is high enough, the neuron turns "on" and sends a signal.

This means a neuron only reacts when the input is strong enough — like a switch that only flips when pushed hard enough.

Learning in Biological Systems

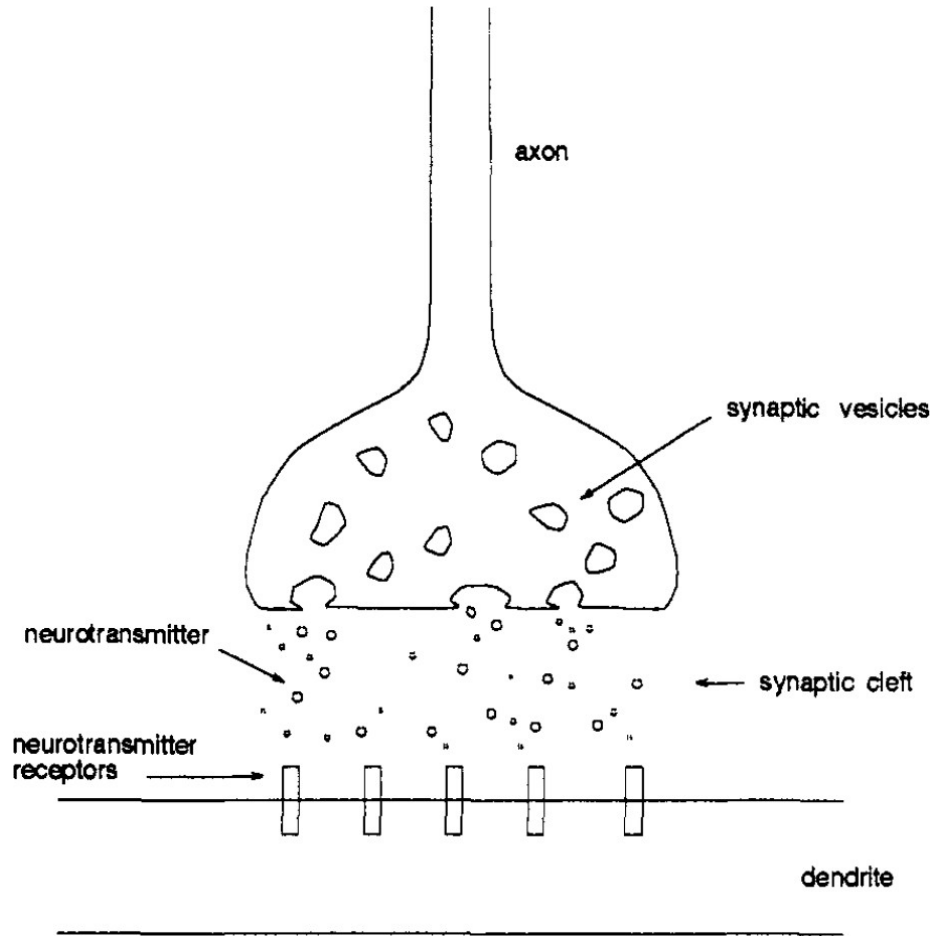
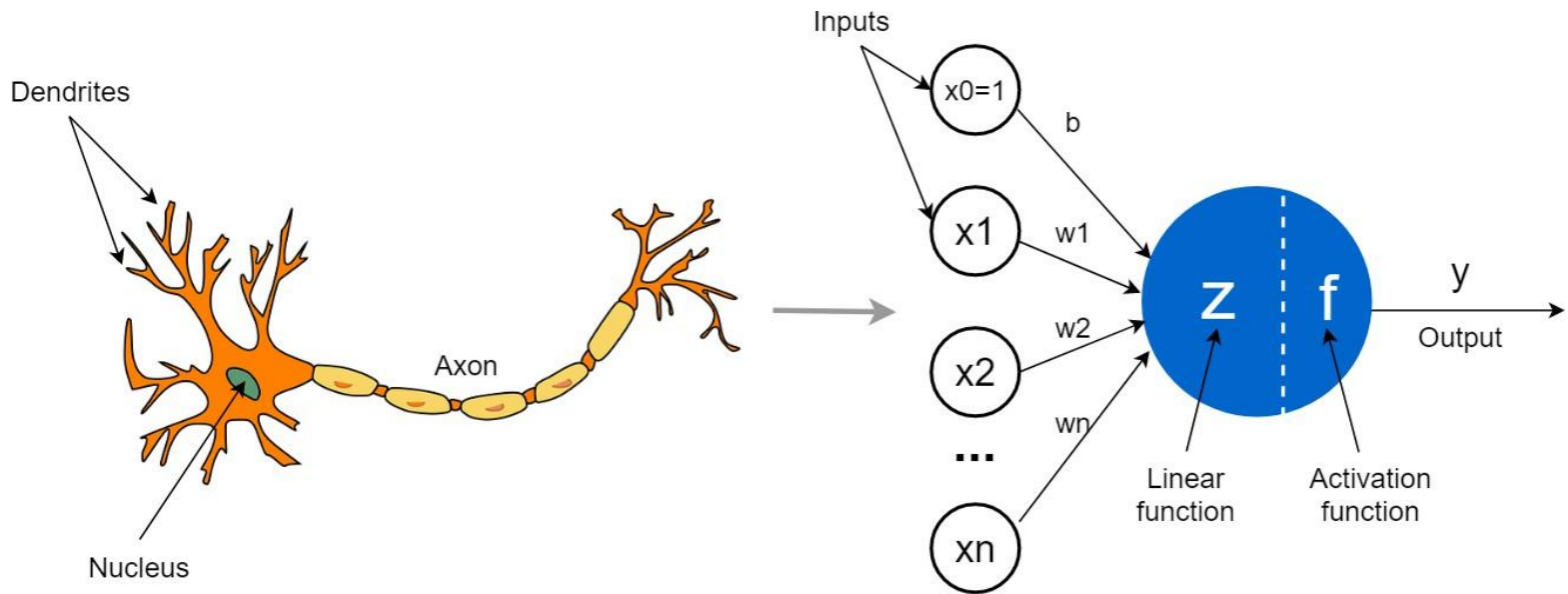


Fig: How Neurons Communicate at the Synapse

When a signal reaches the end of a neuron (the **axon terminal**), it causes small sacs called **synaptic vesicles** to release **neurotransmitters** (chemical messengers). These chemicals travel across a tiny gap called the **synaptic cleft** and bind to **receptors** on the surface of the next neuron's **dendrite**. This triggers the next neuron to continue the signal.



Biological Neuron	Artificial Neuron
Dendrites receive signals from other neurons.	Inputs ($x_1, x_2, \dots, x_n$) receive data or signals.
Nucleus processes the signal inside the cell body.	Linear function (z) processes the weighted sum of inputs.
Axon sends the signal to the next neuron.	Output (y) is passed to the next layer or neuron.
Synapse adjusts signal strength between neurons.	Weights ($w_1, w_2, \dots, w_n$) determine input importance.
Neural firing depends on a threshold.	Activation function (f) decides whether to "fire" (output a value).

Learning In Machines

- ◆ Learning isn't just something living things do — machines can learn too.
- ◆ This challenges the common idea that computers only do what they're told and can't adapt.
- ◆ While it's true that machines follow programs, they can still behave in ways that seem intelligent by learning from experience.
- ◆ To explain this, the section talks about a simple learning machine called **MENACE** (Matchbox Educable Noughts And Crosses Engine), created by **Donald Michie** in the 1960s.
- ◆ MENACE learns to play the game of **noughts and crosses (tic-tac-toe)** using just **matchboxes and beads** — no fancy technology.

Learning In Machines

Here's how it works:

- ◆ MENACE has 288 matchboxes, each representing a different game situation.
- ◆ Each box has colored beads, with each color representing a possible move.
- ◆ To make a move, MENACE randomly picks a bead from the box that matches the current game state.
- ◆ After the game ends, MENACE is updated based on the result:
 - If it wins, more beads of the successful moves are added (rewarding good moves).
 - If it loses, beads for the bad moves are removed (punishing bad choices).
 - If it's a draw, nothing changes.

Learning In Machines

What's special about MENACE:

- ◆ It learns from experience, not instructions.
- ◆ The "knowledge" is not stored in one place, but spread out in the bead choices across all boxes.
- ◆ It solves the big problem (winning the game) by learning from many small situations.
- ◆ Even though each matchbox acts alone, together they help the machine improve.

The Differences

- The brain is excellent at tasks like vision, speech, and learning because it uses many simple parts (neurons) working together in parallel. This structure makes it flexible, able to learn, and tolerant of small failures—if some neurons stop working, others take over. This is called **fault tolerance** and **graceful degradation**.
- In contrast, traditional computers rely on one or a few very fast processors. They are great at simple calculations but are not good at handling complex, varied data like the brain does. If a computer's processor fails, the whole system often crashes. That's why scientists are now trying to build computers that work more like the brain: using simple, connected parts that share the work.

Summary

- ◆ Brain is **parallel, distributed processing system**.
- ◆ Basic processing unit called the **neuron**.
- ◆ Approximately 10^{10} neurons each connected to **10^4** others.
- ◆ **Operation of neuron:** fires pulse down axon when sufficient input received from dendrites. Connections via chemical junctions called synapses.
- ◆ **Learning** increases efficacy of synaptic junction.
- ◆ Machines can learn through **positive reinforcement**.
- ◆ **Cerebral cortex** shows local areas of specialized function.